AIR QUALITY INDEX PREDICTION

Team Signal Seekers.

A Machine Learning Approach for Pollution Classification

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PROBLEM STATEMENT & MOTIVATION

- Air pollution is a serious health concern in major Pakistani cities
- Every winter, Lahore experiences severe smog with reduced visibility
- People suffer from respiratory illnesses due to toxic air
- Need for predictive system to forecast air quality

This project builds a machine learning system that predicts AQI categories from pollutant measurements, enabling real-time air quality monitoring and public health advisories

PROJECT OBJECTIVE

Built ML system to predict AQI categories from pollutant measurements

Input: PM2.5, PM10, NOx, CO, SO2, O3, etc.

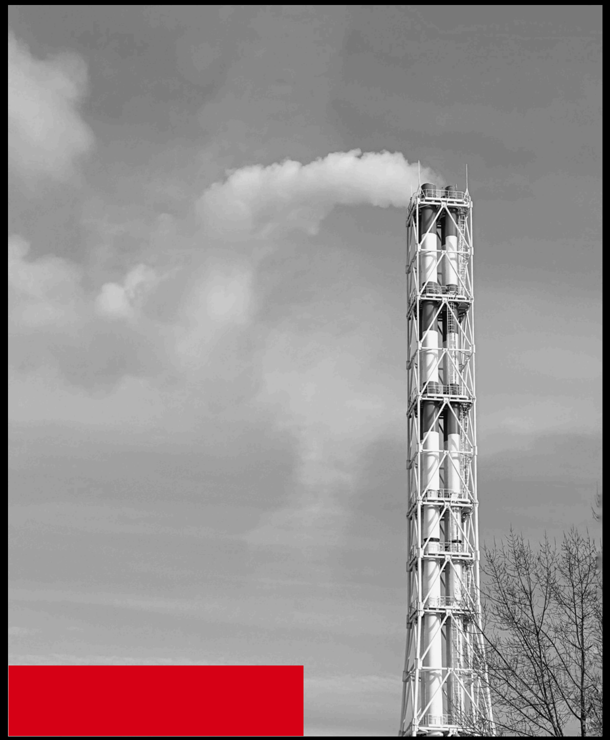
Output: 6 AQI categories (Good → Severe)

Goal: Enable real-time air quality monitoring and public health advisories

DATASET OVERVIEW

- **Source:** Central Pollution Control Board of India (CPCB)
- **Time Period:** 2015-2020 (5 years)
- **Coverage:** 26 Indian cities
- **Original Size:** 29,531 records, 16 columns
- **Final Size:** 24,850 records after cleaning

DATA PREPROCESSING



CLASS	BEFORE	AFTER
Good	1341	7063
Satisfactory	8224	7063
Moderate	8829	7063
Poor	3267	7063
Very Poor	1851	7063
Severe	1338	7063

FEATURE ENGINEERING

Temporal Features

- Month indicators with sine/cosine transformations
- Captures seasonal patterns

Ratio Features

- PM_Ratio (PM2.5/PM10)
- NOx_NO2_Ratio

Interaction Features

- PM_CO_Interaction (PM2.5 × CO)
- NOx_SO2_Interaction

Composite Indices

- Composite_PM (60% PM2.5 + 40% PM10)
- Composite_NOx

EXPLORATORY DATA ANALYSIS

- Significant class imbalance (Moderate & Satisfactory dominate)
- Strong correlations between PM2.5, PM10, and CO (0.68)
- **Clear seasonal pattern:** worse air quality in winter (Nov-Feb)
- **Geographic variation:** Delhi shows many "Severe" episodes

MODELS TESTED

LOGISTIC REGRESSION

RANDOM FOREST

SUPPORT VECTOR MACHINE (SVM)

MULTI-LAYER PERCEPTRON (MLP)

MODEL 1 - LOGISTIC REGRESSION

71.93 %

Accuracy

0.7264

F1 Score

Type: Linear Classifier

Hyperparameters: C=1.5, multinomial, balanced weights

MODEL 2 - RANDOM FOREST

80.82 %

Accuracy

0.8092

F1 Score

Type: Linear Ensemble of 300 decision trees

Hyperparameters: max_depth=20, min_samples_split=5

MODEL 3 - SUPPORT VECTOR MACHINE

74.65 %

Accuracy

0.7527

F1 Score

Type: RBF kernel classifier

Hyperparameters: C=1.0, gamma='scale'

MODEL 4 - MLP NEURAL NETWORK

76.68 %

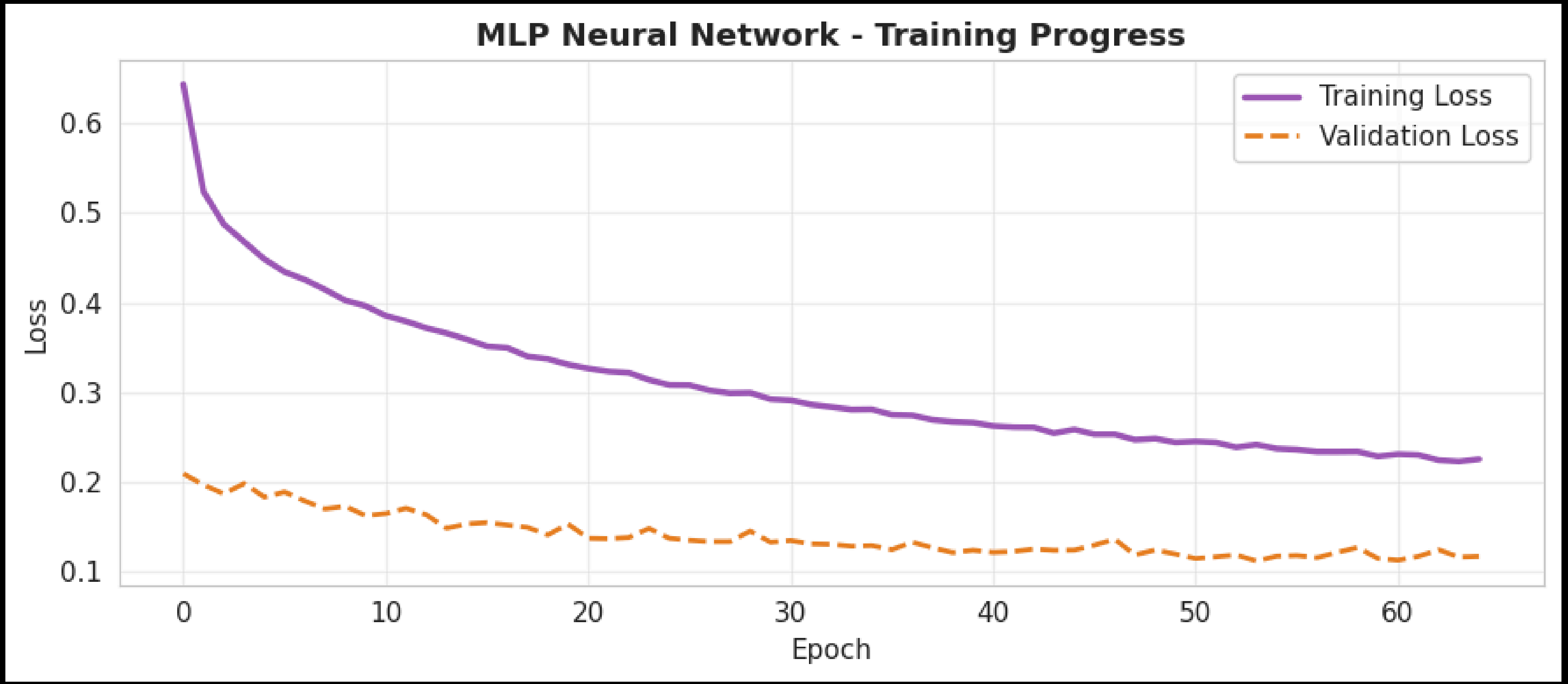
Accuracy

0.7692

F1 Score

Architecture: Input(21) → 128 → 64 → 32 → Output(6)

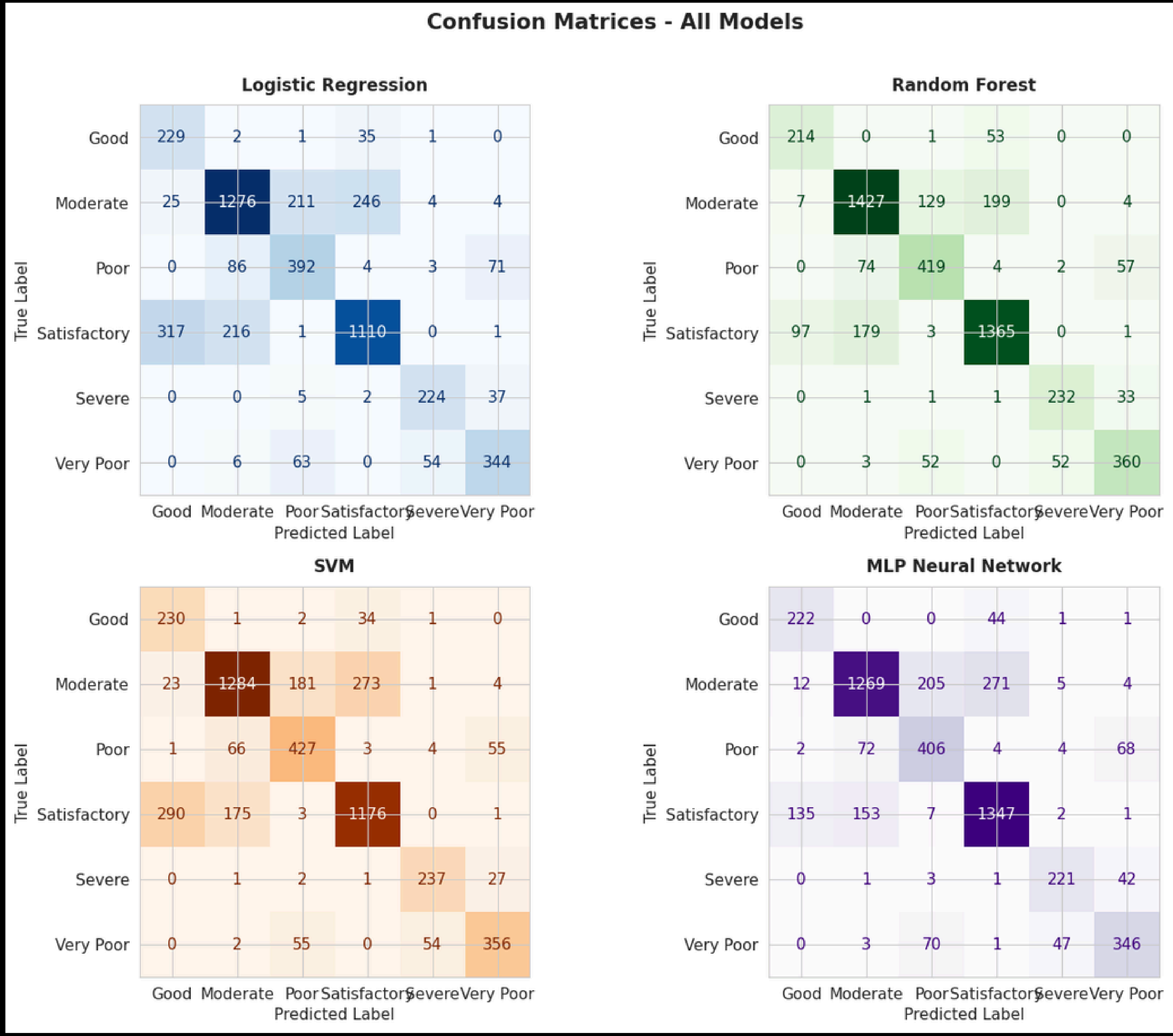
Hyperparameters: ReLU activation, batch_size=32, early stopping



MODEL PERFORMANCE COMPARISON

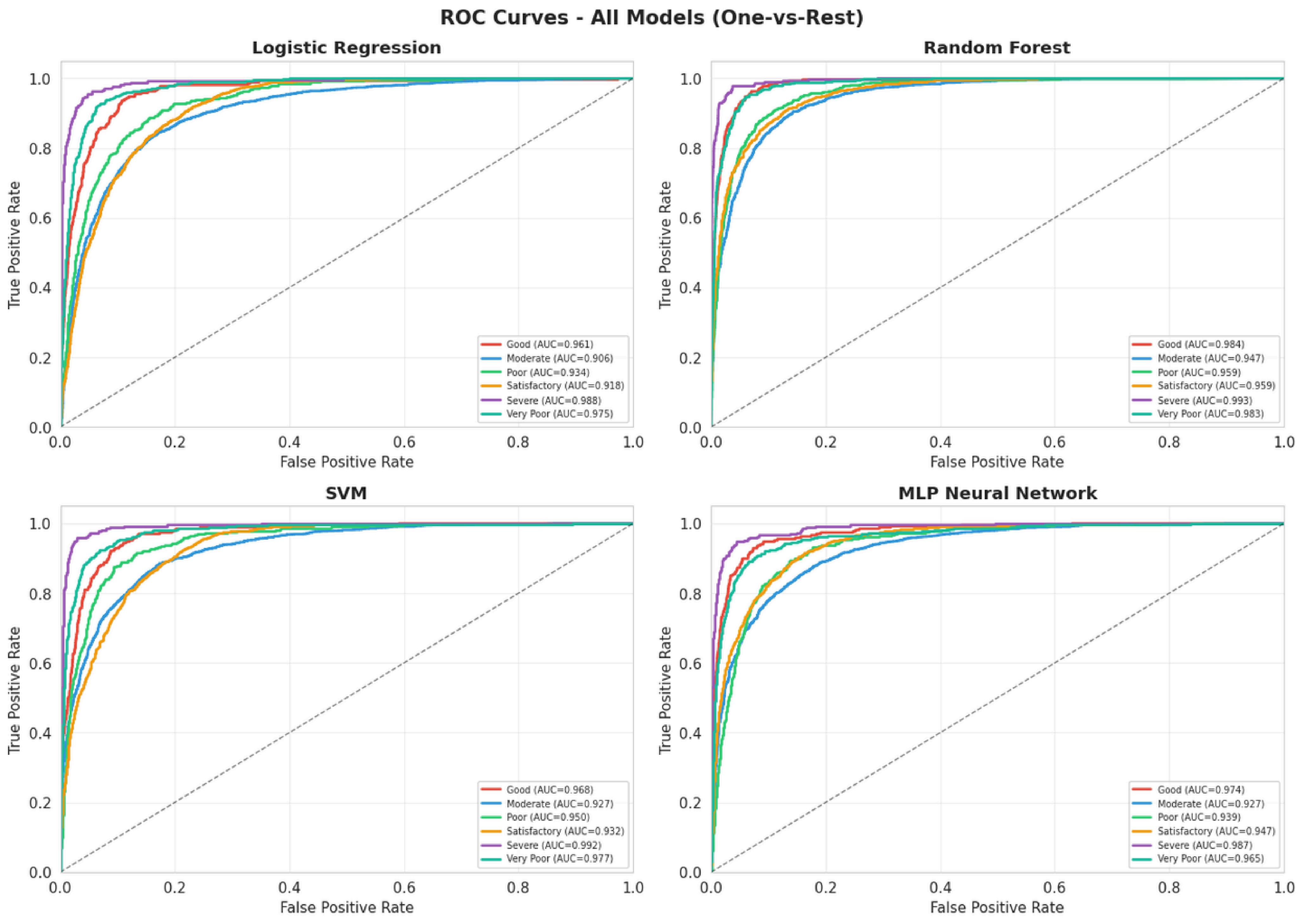
CONFUSION MATRIX ANALYSIS

- Most misclassifications occur between adjacent categories
- Random Forest has strongest diagonal (correct predictions)
- Excellent performance on critical "Severe" and "Very Poor" categories



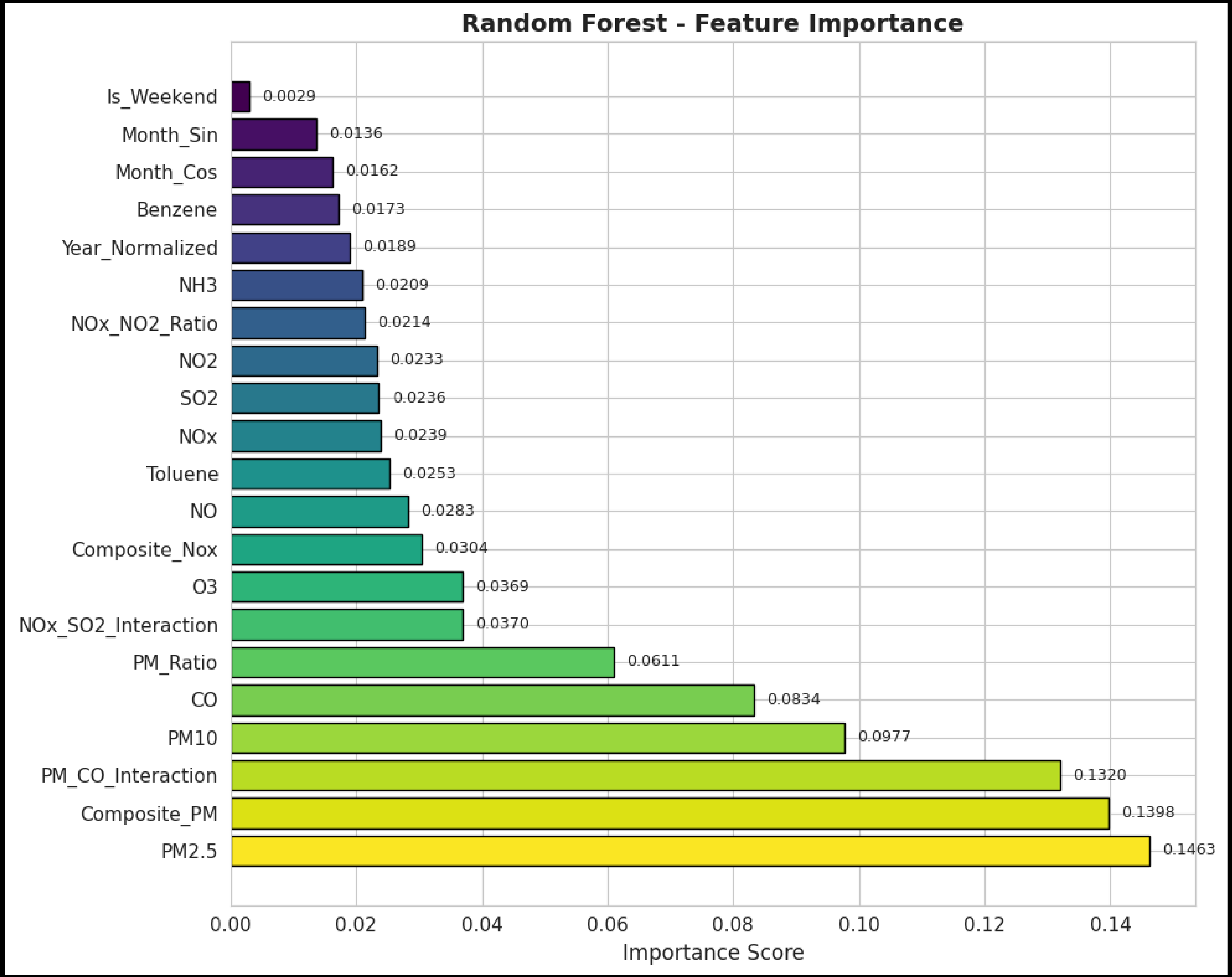
ROC CURVE ANALYSIS

- Random Forest and MLP show exceptional discriminative capacity
- AUC values exceed 0.95 for critical categories
- Model can reliably distinguish between risk levels



FEATURE IMPORTANCE

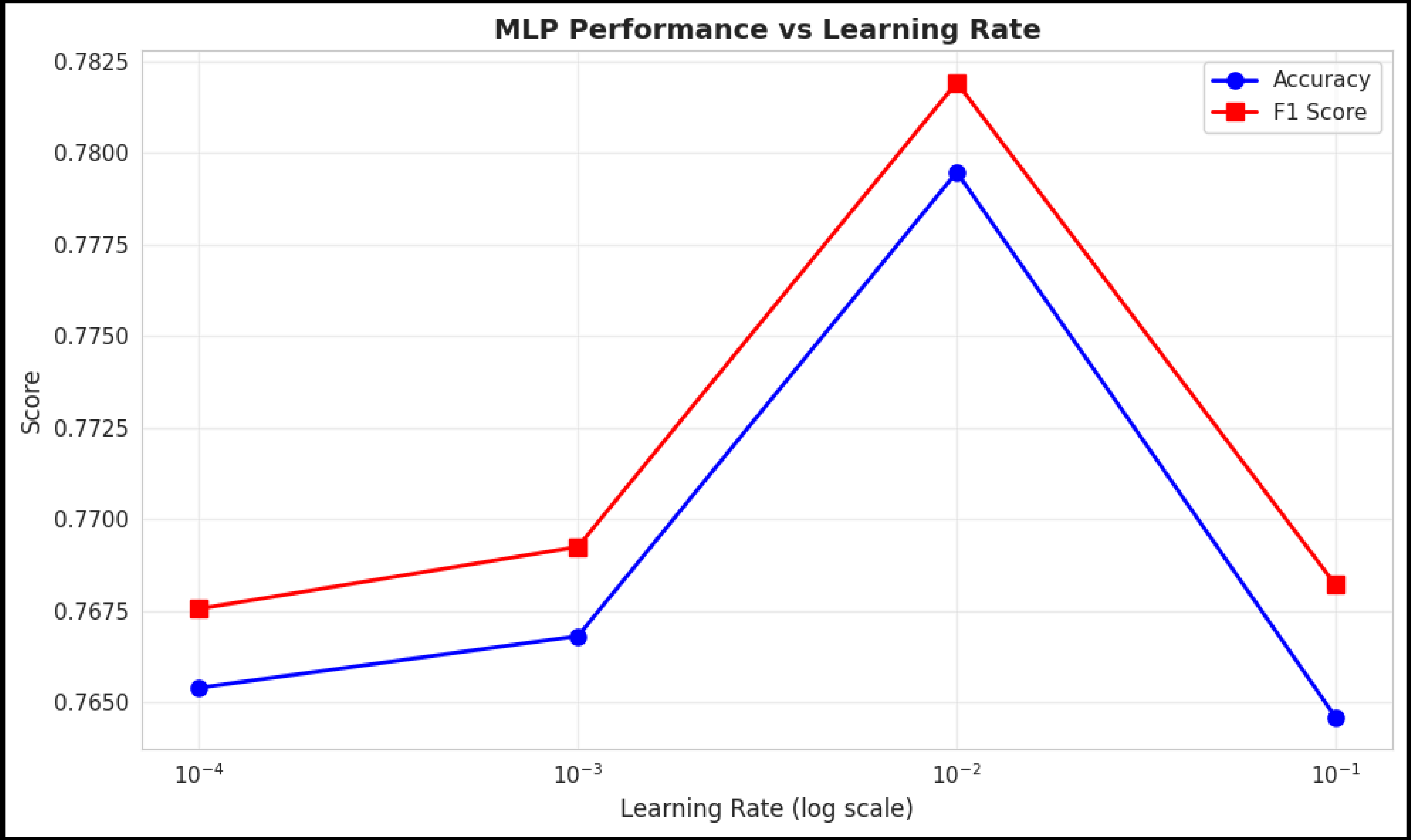
- PM2.5 - 0.1463 (Most important!)
- Composite_PM - 0.1398
- PM_CO_Interaction - 0.1320
- PM10 - 0.0977
- CO - 0.0834



MODEL STABILITY - EXPERIMENT 1

Learning Rate Sensitivity (MLP)

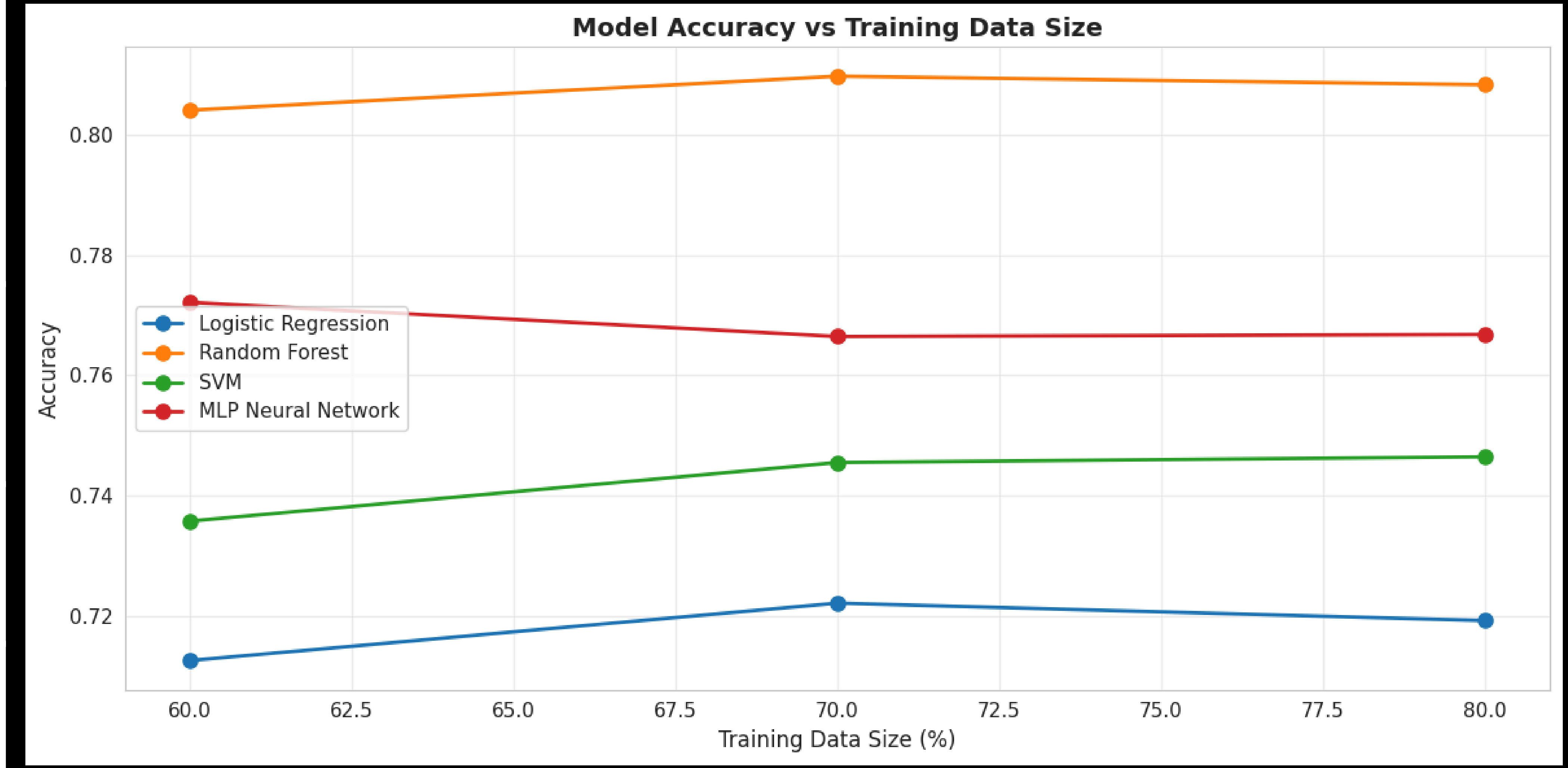
Learning Rate	Accuracy	F1 Score
0.1	72.34%	0.7289
0.01	75.12%	0.7543
0.001	76.68%	0.7692
0.0001	75.89%	0.7618



MODEL STABILITY - EXPERIMENT 2

Train-Test Split Sensitivity

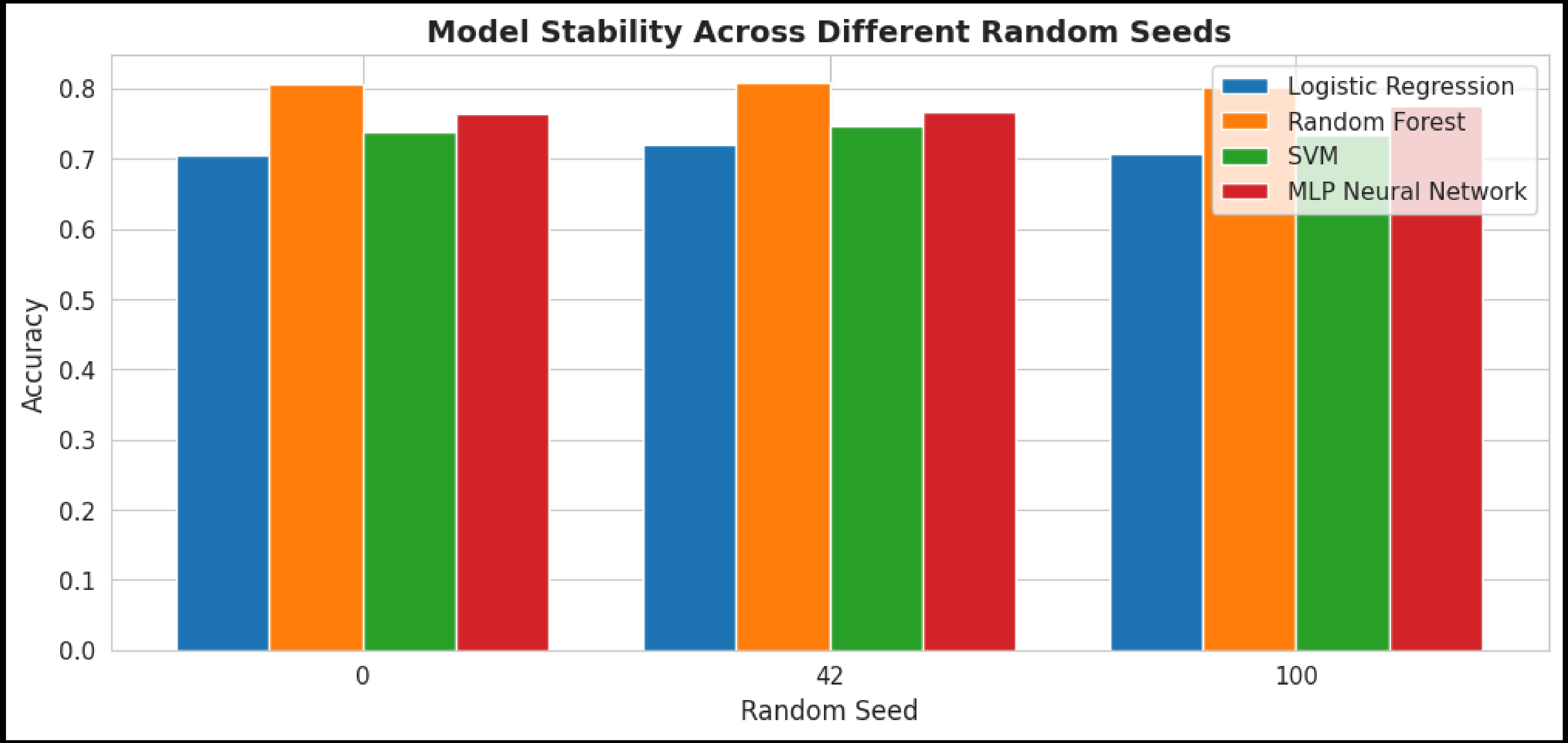
Model	80 - 20	70 - 30	60 - 40
Logistic Regression	71.93 %	71.24 %	70.56 %
Random Forest	80.82 %	80.15 %	79.43%
SVM	74.65 %	73.98 %	73.12 %
MLP Neural Network	76.68 %	76.01 %	75.27 %



MODEL STABILITY - EXPERIMENT 3

Random Seed Stability

- **Logistic Regression:** 0.0002
- **Random Forest:** 0.0001
- **SVM:** 0.0002
- **MLP:** 0.0002



REAL WORLD TESTING

Scenario	Predicted AQI
Clean Air (Mountain/Forest)	Good
Moderate Pollution (Small City)	Satisfactory
High Pollution (Industrial/Winter)	Very Poor
Severe Pollution (Smog/Diwali)	Severe

KEY FINDINGS - SUMMARY

Best Model

Random Forest

Most Important Features

PM2.5 (0.1463), Composite_PM (0.1398)

High Stability

Std dev < 0.0003 across random seeds

Optimal Learning Rate

0.001 for MLP

Seasonal Pattern

Worse air quality in winter months

Strong Generalization

Perfect real-world scenario predictions

LIMITATIONS

- Dataset limited to Indian cities (not Pakistani data)
- No meteorological factors included (temperature, humidity, wind)
- Classification only (not continuous AQI values)
- 5-year-old data (2015-2020)

FUTURE WORK

- Incorporate weather/meteorological data
- Hyperparameter optimization with GridSearchCV
- Deploy as web API for public access
- Test on Pakistani cities (Lahore, Karachi, Islamabad)
- Experiment with LSTM/GRU for time-series forecasting

CONCLUSION

Team Signal Seekers:

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- Mahira Ali
- Kinza Choudhry